



Gamified Microlearning Platform for Technical Skills

Submitted by: Rojin Singh Mahat

Student id: 2417733

Group: L6CG10

Supervisor name: MD Saddam Khan

Supervisor code: 134

Date: 2025/11/27

Contents

Abstract	1
Introduction.....	2
Problem Statement	3
Objectives	4
Related works/Existing systems	5
Literature Review.....	7
Methodology	8
1. Research Approach	8
2. Development Methodology.....	8
3. System Design & Tools.....	8
4. Evaluation Strategy	8
5. Development Workflow.....	9
System architecture and Diagrams.....	10
Functional Decomposition Diagram (FDD) & Functionalities	10
Use Case Diagram	11
Activity Diagram.....	12
Sequence Diagram	13
Expected Outcomes	15
Project Scope	16
Project Management	17
Methodology:.....	17
Tools & Tracking:.....	17
Gantt Chart & Timeline:	17
Risk Management	19
Risk Register (Narrative Format).....	19
Limitations	23
Conclusion & Future Scope	24

Conclusion:.....	24
Future Scope:	24
Research Articles & Studies	25
Tools, Libraries, & Platforms	27

Abstract

Workplaces today require people to constantly pick up new skills, but most training platforms are long, boring, and hard to stick with. Our project focuses on solving that problem by building a gamified learning app that teaches technical micro-skills. They are small but useful skills like SQL basics, networking fundamentals, Excel formulas, and cybersecurity awareness. These are the kinds of skills that actually make a difference in daily work but are usually overlooked or taught in outdated ways.

The system will use gamification (points, levels, streaks, badges, sounds, and animations) to keep learners engaged while delivering short, focused lessons that can be completed in minutes. This makes learning feel less like a chore and more like a challenge. The app is designed to be modular, meaning we can easily add new micro-skills over time without changing the core system.

My main goal is to prove that gamified microlearning can increase engagement and retention compared to traditional training methods, while also producing a platform that feels polished and ready to use.

Introduction

In today's fast-evolving technology landscape, continuous learning is essential for both students and professionals. Traditional education often struggles to provide hands-on, project-based experience, leaving learners underprepared for real-world scenarios. Modern online learning platforms, such as Boot.dev, freeCodeCamp, Codecademy, and Exercism, have emerged to bridge this gap by offering interactive, structured, and practice-oriented courses. Other notable platforms, including Codio, BigDevSoon, GoSkills, and Kodeco, emphasize gamified microlearning, cloud-based development environments, and real-world project simulations to enhance skill acquisition and retention.

Despite the variety of platforms, there remains a need for a system that combines micro-skill modules, gamified progression, and workplace-oriented training in a single integrated environment. This project aims to address this gap by developing a platform that delivers bite-sized, gamified technical learning experiences, ensuring learners not only acquire skills efficiently but also remain motivated throughout the process.

Problem Statement

Traditional e-learning platforms have grown significantly in recent years, offering flexibility and accessibility to learners worldwide. However, many of these platforms still struggle with sustained learner engagement, motivation, and completion rates. According to Sailer and Homner (2020), while gamification has shown potential in enhancing learning outcomes, its impact varies significantly depending on the design and integration of game elements. Similarly, Landers and Armstrong (2017) argue that gamification alone does not guarantee better performance unless carefully aligned with instructional objectives, highlighting the risk of superficial or ineffective implementations.

Furthermore, recent studies emphasize that personalization and adaptive learning remain underdeveloped in existing platforms. Alhammad and Moreno (2018) note that most gamified systems fail to account for individual learner differences, which often leads to uneven learning outcomes. More recent work, such as Kumar et al. (2024), reinforces this by pointing out that engagement strategies must evolve with learner needs, particularly for digital-native generations who demand interactive and motivating experiences.

Another challenge lies in balancing scalability with meaningful interactivity. For instance, Caballé and Clarisó (2014) highlight that many large-scale learning systems struggle to provide both engagement and pedagogical depth, often sacrificing one for the other. Meanwhile, Alhammadi et al. (2023) demonstrate that while AI-powered adaptivity can enhance personalized learning, there is still limited practical application in gamified technical skill training platforms.

Therefore, the problem this project addresses is the lack of an accessible, engaging, and adaptive gamified learning system tailored for micro-technical skills (e.g., SQL basics, networking fundamentals, Excel formulas). While existing platforms such as Coursera, Udemy, and Boot.dev provide technical learning opportunities, they often lack gamified progression systems that sustain learner motivation and adaptive personalization that accommodates different learning styles. This gap presents an opportunity to design a platform that combines evidence-based gamification with micro-learning to enhance skill acquisition and retention.

Objectives

The primary objective of this project is to design and develop a gamified web-based learning platform that teaches technical micro-skills essential in modern work environments. To achieve this, the project sets the following specific objectives:

1. Develop a gamified learning environment that incorporates elements such as points, badges, levels, and progress tracking to improve learner motivation and engagement.
2. Focus on technical micro-skills such as SQL basics, Excel formulas, networking fundamentals, and cybersecurity hygiene, providing learners with short, targeted modules.
3. Integrate interactive animations and sound-based feedback to enhance the learning experience and simulate real-time skill application.
4. Ensure accessibility and usability by designing an intuitive user interface that caters to learners with different levels of technical background.
5. Evaluate effectiveness by comparing learner engagement and knowledge retention with traditional non-gamified methods, drawing insights from prior studies (e.g., Sailer and Homner, 2020; Kumar et al., 2024).
6. Lay groundwork for scalability, enabling future integration of additional micro-skills or mobile support.

Related works/Existing systems

There are a number of similar applications and web platforms which provide similar features like mine. While they are good and even successful, they still lack some things I aim to cover. Some of such platforms are as follows:

- **Khan academy / freeCodeCamp / CodeCademy:**
 - They teach coding and technical skills online using modules and quizzes
 - **Strengths:**
 - ♦ Structured curriculum
 - ♦ Immediate feedback
 - ♦ Some gamification elements
 - **Weaknesses:**
 - ♦ Lightly gamified
 - ♦ Micro-skill focus is broad
 - ♦ Modules are not bite sized
 - ♦ No deep tracking of incremental skill mastery
- **Duolingo (for general learning gamification)**
 - They gamify learning a new language via points, streaks, levels, leaderboards and are the biggest free language learning app
 - **Strengths:**
 - ♦ It is highly engaging
 - ♦ It shows clear progression of learner's skills
 - ♦ Gives rewards to keep users hooked
 - **Weaknesses:**
 - ♦ Doesn't specifically teach technical skills
 - ♦ Lacks real-world applications
- **Sololearn (tech skill apps)**
 - They are a mobile-first application that provide micro coding lessons, exercises and instant feedback
 - **Strengths:**
 - ♦ Short lessons
 - ♦ Mobile convenience
 - ♦ They have a social community
 - **Weaknesses:**
 - ♦ Limited adaptive learning
 - ♦ Shallow skill mastery tracking
 - ♦ Minimal gamification compared Duolingo-styled apps

While existing platforms like Khan Academy, Duolingo, and SoloLearn provide valuable learning experiences, they either lack a focus on bite-sized technical micro-skills, or their gamification elements are superficial. Our system addresses these gaps by combining micro-skill-based lessons with adaptive gamified elements that track and enhance incremental skill mastery.

Literature Review

Gamification in education has been a growing research area, with several studies demonstrating its potential to improve engagement and learning outcomes. However, the application of gamification to the acquisition of micro-skills, particularly in technical fields, remains underexplored. This review synthesizes five key studies relevant to the proposed project.

Caponetto et al. (2014) conducted one of the earlier systematic explorations of gamification in education, noting that while gamified approaches increased learner motivation, the implementations were often fragmented and lacked consistency. This highlights the need for structured systems that integrate both game mechanics and pedagogical design. Similarly, Hamari, Koivisto and Sarsa (2014) in their meta-analysis found mixed results, concluding that gamification's effectiveness depends strongly on context, design, and learner profiles. These findings underline the importance of designing gamification tailored to the learning of specific micro-skills.

More recently, Wang et al. (2018) examined how gamified mobile platforms affected student collaboration and performance. Their study demonstrated that gamified interventions significantly increased engagement and peer interaction, but also cautioned that poor alignment between the gamified elements and the learning objectives could reduce effectiveness. This aligns directly with the current project's aim of carefully aligning micro-skill development with meaningful gamified mechanics.

Huang et al. (2023) explored gamification through the lens of online higher education, finding that personalized gamified elements improved not only engagement but also persistence in learners. They emphasized adaptive systems that account for individual differences, an insight that can be applied to tailoring micro-skill training for diverse learners. Similarly, Salas-Morera et al. (2024) reported that gamification improved both short-term learning and long-term retention in technical subjects, suggesting its suitability for training micro-skills that require practice and reinforcement.

Finally, Chen et al. (2024) investigated the use of gamification in large-scale online learning environments. Their findings emphasized scalability and inclusivity, with gamified interventions leading to higher course completion rates and broader learner participation across demographics. This provides evidence that gamified micro-skill learning platforms can potentially reach wide audiences without sacrificing effectiveness.

In summary, the literature demonstrates that gamification enhances engagement and retention across diverse contexts, but effectiveness depends on proper alignment with learning goals, adaptive design, and scalability. Current research has yet to fully explore gamification applied to micro-skills training in technical domains, which represents the gap that this project seeks to address.

Methodology

1. Research Approach

This project will adopt a design- and development-focused approach, informed by existing literature on gamification and microlearning. The focus will be on implementing a functional prototype that demonstrates how gamified elements can enhance engagement in technical micro-skill learning. Informal user feedback will be collected during prototype demonstrations to identify usability issues and gather qualitative insights on user experience.

2. Development Methodology

The system will be developed using an Agile methodology, specifically a Scrum framework with two-week sprints. Agile was chosen for its flexibility and iterative improvement capabilities. Each sprint will deliver a functional increment of the system, such as authentication, gamification engine, micro-skill modules, and multimedia integration. Task management and progress tracking will be handled using JIRA or Trello, ensuring transparency and accountability in development.

3. System Design & Tools

- **Frontend Development:** React.js will be used for responsive and component-based UI design. Animations will be integrated using Framer Motion and Lottie for smooth micro-interactions.
- **Backend Development:** Node.js with Express.js will provide a scalable backend, with MongoDB storing user data, progress, and learning history.
- **Gamification Engine:** Core elements (XP, levels, badges, leaderboards) will be implemented as configurable modules in Node.js, allowing easy addition of new micro-skills.
- **Game/Animation Layer:** Interactive learning modules will utilize Phaser.js for HTML5 games. Unity WebGL may be integrated for advanced animated modules.
- **Design & Assets:** Wireframes and prototypes will be created in Figma. Visual assets will be developed using Adobe Illustrator and Blender. Free-to-use sound effects and background music will be sourced from [Freesound.org](https://freesound.org) and [OpenGameArt.org](https://opengameart.org).
- **Version Control:** GitHub/GitLab will manage the codebase for collaboration and version tracking.

4. Evaluation Strategy

The prototype will undergo informal usability testing and feedback sessions with a small group

of volunteer participants. The aim is to identify usability issues, gather opinions on gamification engagement, and ensure the platform is intuitive and motivating. Formal quantitative data collection is not included in this phase; instead, evaluation focuses on practical demonstration of the system's gamification features and learning module design. Future work could expand to formal studies on engagement and retention.

5. Development Workflow

1. **Requirement Analysis & Research** – identify micro-skills, gamification features, and pedagogical grounding.
2. **System Design** – wireframing, prototype design, asset creation.
3. **Core Development** – implement frontend, backend, and gamification mechanics.
4. **Integration** – animations, sounds, and interactive elements.
5. **Testing & Iteration** – Unit testing, integration testing, user acceptance testing using tools like postman, jest
6. **Deployment** – host the final system on a cloud platform such as AWS or Heroku.

System architecture and Diagrams

Functional Decomposition Diagram (FDD) & Functionalities

1. User Management

- Register new learner accounts
- Authenticate existing users via login
- Manage user sessions (log out, session handling)
- Profile management (view/update basic info)

2. Lesson Management

- Browse catalog of microskills
- Select lesson difficulty (Beginner, Intermediate, Advanced)
- Access lesson content (modules, quizzes, exercises)
- Track lesson completion status

3. Gamification & Rewards

- Earn points upon successful lesson completion
- Unlock badges and levels based on milestones
- Maintain streaks for consistent learning
- Display progress dashboard

4. Assessment & Feedback

- Attempt quizzes/interactive exercises
- Submit answers for evaluation
- Receive instant feedback (correct/incorrect, hints)
- Retry failed questions/lessons

5. Progress Tracking

- Store learner activity history
- Update and display overall progress

- Generate simple performance summaries

6. Admin/Instructor Management

- Add, update, and remove lesson modules
- Set lesson difficulty levels
- Monitor learner statistics and engagement
- Restrict/unlock access based on system rules

7. Payment & Monetization

- Enable paid access to higher-level content
- Handle transactions for premium lessons

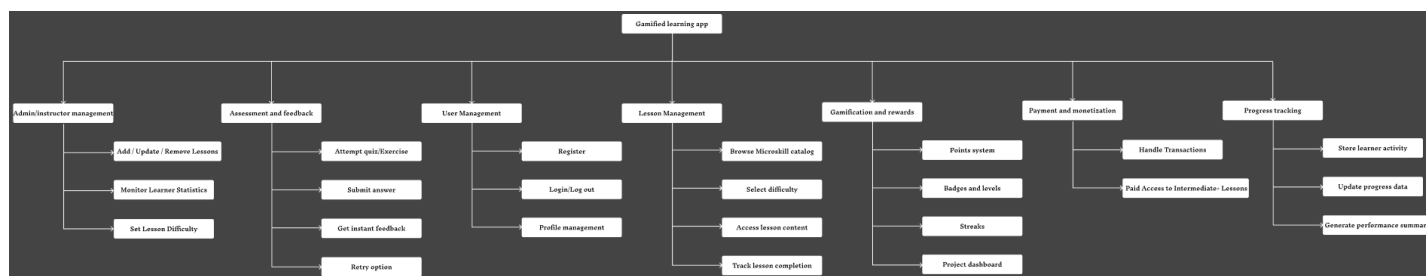


FIG: FUNCTIONAL DECOMPOSITION DIAGRAM

Use Case Diagram

The Use Case Diagram illustrates the main interactions between the system and its stakeholders: learners, administrators/instructors, and the platform itself. Learners are able to register, log in, browse microskill lessons, select a difficulty level, attempt lessons and quizzes, receive feedback, and earn rewards. Administrators and instructors manage the creation and structuring of lessons, define difficulty levels, and monitor learner performance to ensure effective delivery of content. The system automatically handles processes such as progress tracking and reward distribution.

This diagram is important because it defines the functional boundaries of the system and shows, at a high level, which actions are initiated by users and which are handled internally. It provides a foundation for understanding how different actors will engage with the platform.

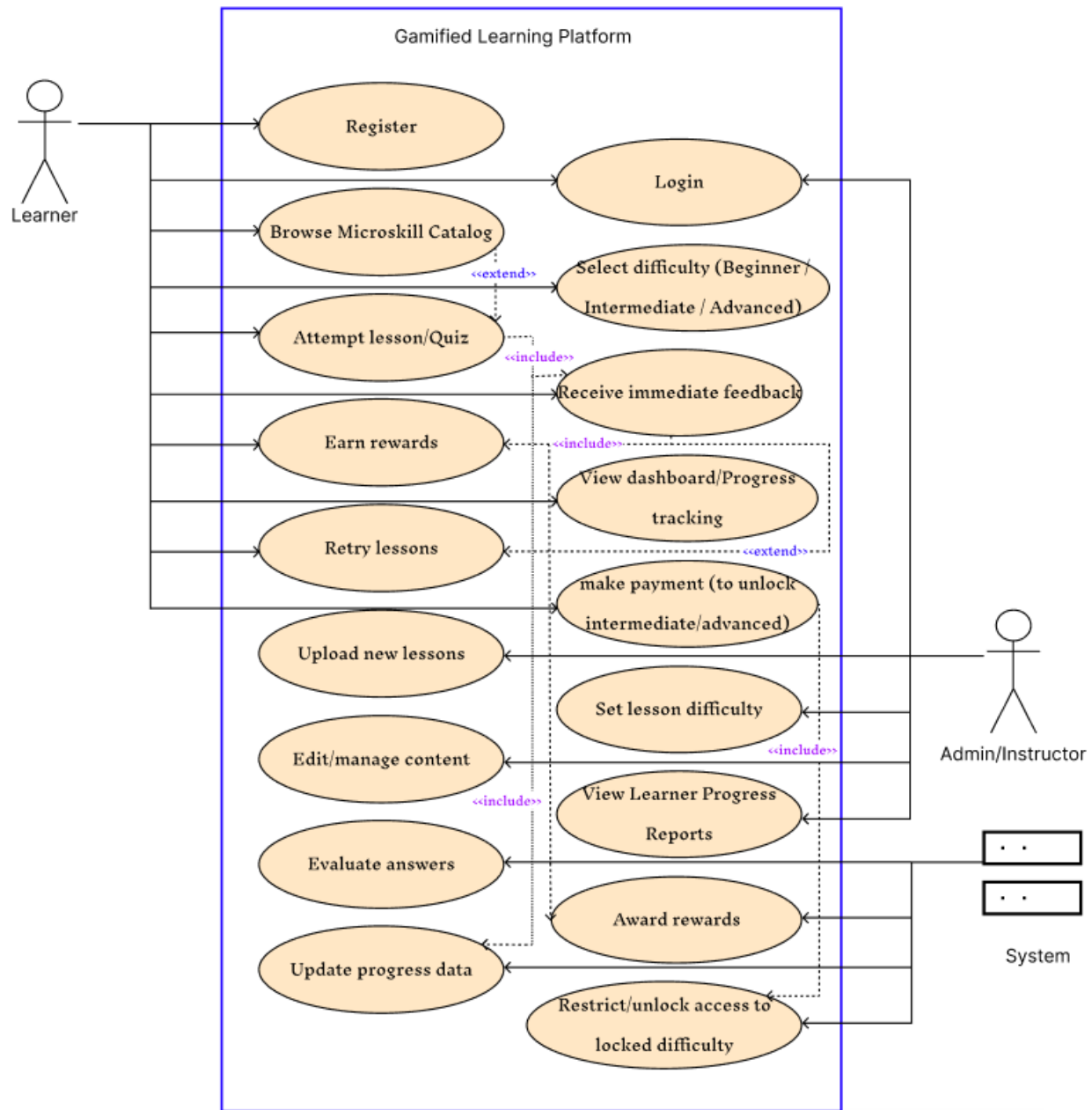


FIG: USE CASE DIAGRAM

Activity Diagram

The Activity Diagram describes the workflow of a learner navigating and using the system. The process begins with the learner logging in and selecting a desired lesson difficulty. From there, the learner attempts a lesson or quiz, submits answers, and receives immediate evaluation. Based on the result, the flow diverges: a correct answer leads to rewards and updated progress, while an incorrect answer leads to feedback and the option to retry. Administrators also have activities related to managing lesson content and monitoring learner data, while the system automates

tracking and updates throughout the process.

This diagram is significant because it captures the dynamic behavior of the system, emphasizing decision points and parallel flows. It helps clarify how user actions and system responses interconnect to create a seamless learning experience.

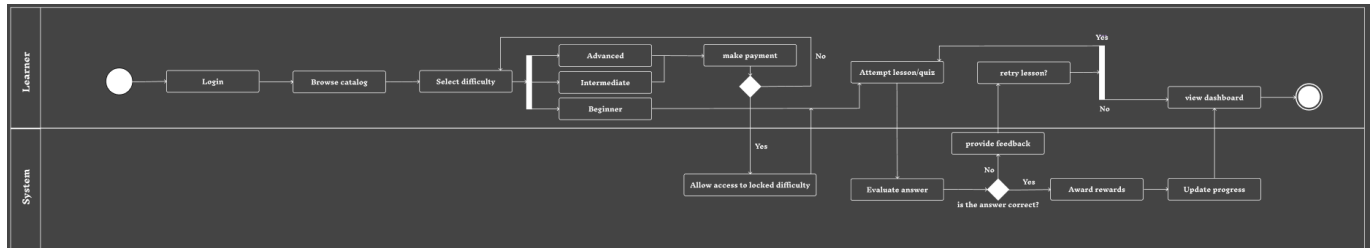


FIG: ACTIVITY DIAGRAM

Sequence Diagram

The Sequence Diagram details the chronological interactions between the system components during the learner's journey. When a learner logs in, the Platform UI communicates with the Database to validate credentials before displaying the dashboard. Upon selecting a difficulty and starting a lesson, the Lesson Module retrieves lesson data from the Database and delivers it to the learner. During answer submission, the Evaluation Engine checks the learner's response against stored answers and returns results. If the answer is correct, the Reward System is triggered to update learner rewards and progress records; if incorrect, the learner receives feedback and retry options.

This diagram is crucial because it demonstrates how system components collaborate in real time, highlighting both synchronous and asynchronous communication. It ensures clarity on how data flows between the UI, database, and backend modules, which is essential for system design and later implementation.

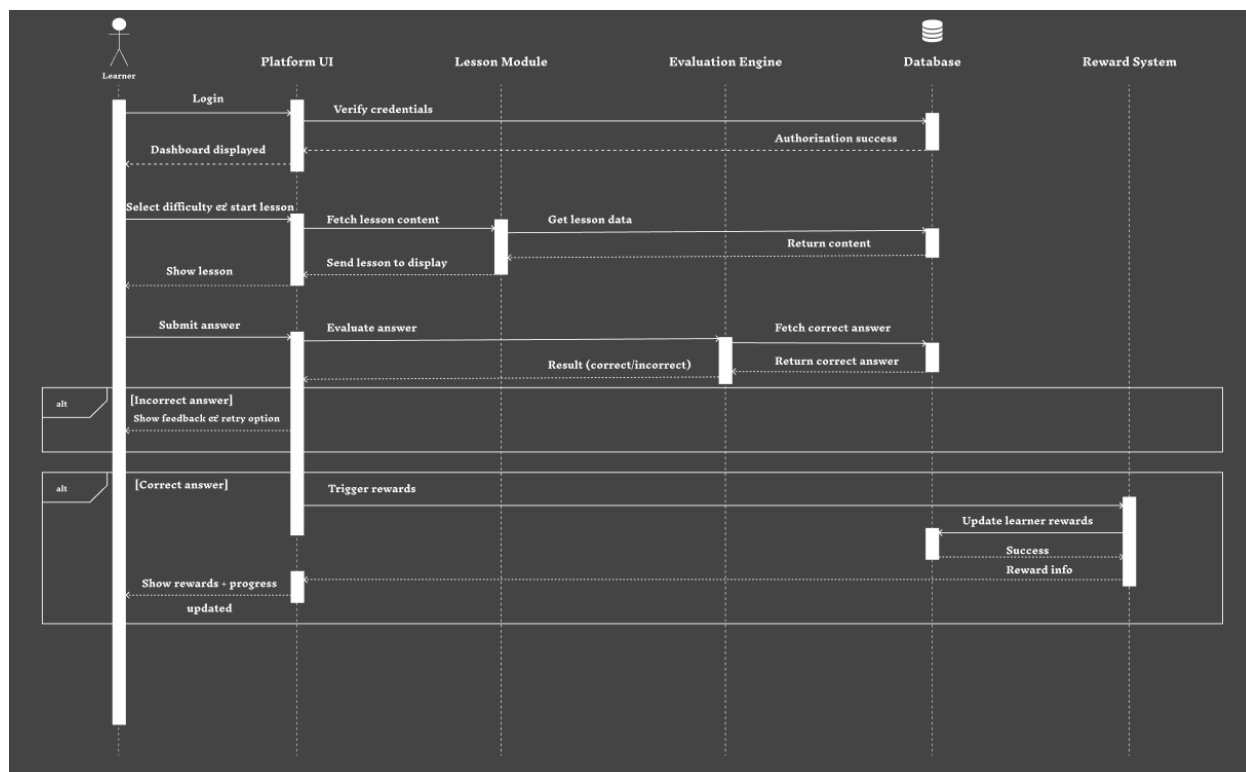


FIG: SEQUENCE DIAGRAM

Expected Outcomes

The project is expected to deliver a functional **gamified learning platform** for technical micro-skills with the following outcomes:

1. **Boosted Engagement:**

Gamification elements such as points, badges, levels, and streaks will encourage learners to interact with the platform and complete modules, creating a more motivating experience (Hosseini et al., 2018; Papamitsiou & Economides, 2014).

2. **Improved Knowledge Retention (Qualitative):**

Through structured, bite-sized challenges and progressive modules, learners are expected to retain and apply micro-skills more effectively, based on observed engagement and feedback (Gupta et al., 2018; Khamparia et al., 2018).

3. **Intuitive and Accessible Experience:**

The platform will prioritize clear navigation, responsive design, and accessibility, ensuring learners of varying technical backgrounds can interact with the system smoothly (Arumugam & Suguna, 2019).

4. **Demonstrable Prototype for Microlearning:**

A working web-based prototype will serve as proof-of-concept, showcasing the integration of gamification elements, interactive modules, and modular micro-skill content.

5. **Foundation for Future Expansion:**

The system will be designed modularly, allowing new micro-skills, animations, or gamified features to be added easily in future iterations, without major rework.

Project Scope

This project focuses on designing and developing a web-based gamified learning platform for technical micro-skills. The main scope includes:

1. Core Features:

- Gamified progression (points, badges, levels, streaks)
- Modular micro-skill lessons (e.g., SQL basics, Excel formulas, networking fundamentals)
- Interactive animations and sound feedback to enhance engagement
- Intuitive and responsive user interface

2. Excluded Features (Out of Scope):

- Mobile app development (future expansion possible)
- Advanced AI-driven personalization and adaptive learning
- Formal analytics or large-scale user testing

3. Deliverables:

- A functional web-based prototype demonstrating gamified microlearning
- Demonstration of modular content addition
- Documented design and architecture for future iterations

Project Management

Methodology:

Agile Scrum is chosen for flexibility, iterative development, and rapid feedback integration. The development will proceed in **two-week sprints** with specific deliverables at each stage. MERN stack was selected for its scalability, active community support, and industry relevance. React provides component reusability, Node.js enables efficient real-time features, and MongoDB offers flexible schema design suitable for evolving gamification features.

Tools & Tracking:

- **Task Management:** Trello/JIRA for sprint planning, backlog management, and progress tracking
- **Version Control:** GitHub/GitLab for source control and collaboration
- **Design & Prototyping:** Figma for wireframes, Adobe Illustrator/Blender for visual assets
- **Testing Strategy:**
 - **Unit Testing:** Jest for backend functions
 - **API Testing:** Postman for endpoint validation
 - **Integration Testing:** Frontend-backend connectivity
 - **User Testing:** Small group feedback (5-10 users)
 - **Performance Testing:** Load testing with simulated users

Gantt Chart & Timeline:

The project was planned and executed following the Scrum agile methodology, with clearly defined sprints for iterative development. Tasks were prioritized to first deliver the core MVP, followed by additional features, and finally system stabilization and testing. The Gantt chart reflects dependencies between tasks and phases, ensuring a launch-ready platform is completed by the FYP submission deadline. This structured timeline allowed for continuous monitoring of progress and adjustment of priorities as needed throughout the development cycle.

Phases	Sprints	Tasks	Start Date	End Date	Duration
Phase 0: Setup and planning	Sprint 0	Project repo setup, dev environment	9/13/2025	9/26/2025	13
		Database schema planning			
		Hosting/service selection (Vercel, Heroku, etc.)			
		Sprint 1 planning + backlog creation			
Phase 1: Core MVP	Sprint 1	User authentication (register/login)	9/27/2025	10/10/2025	13
		Basic dashboard layout			
		Create lesson module framework			
	Sprint 2	Browse micro-skill catalog	10/11/2025	10/24/2025	13
		Display lessons, basic content			
		Start attempt lesson/quiz flow (stubbed answers)			
	Sprint 3	Immediate feedback system	10/25/2025	11/7/2025	13
		Points & badges for completed lessons			
		Track basic learner progress			
	Sprint 4	Retry lessons option	11/8/2025	11/21/2025	13
		Leaderboard placeholder (optional for MVP)			
		Bug fixes & polish for MVP			
Phase 2: Feature completion	Sprint 5	Select difficulty integration	11/22/2025	12/5/2025	13
		Unlock/lock logic for lessons			
		Payment flow stub (for intermediate/paid levels)			
	Sprint 6	Admin panel: set difficulty, manage lessons	12/6/2025	12/19/2025	13
		Progress tracking dashboard for admins			
		Reward automation linked to lessons			
	Sprint 7	Payment gateway integration (sandbox)	12/20/2025	1/2/2026	13
		Ensure conditional access based on difficulty/payment			
		Fix issues in core flows			
	Sprint 8	Interactive feedback (animations, sounds)	1/3/2026	1/16/2026	13
		UI/UX polish for learner dashboard			
		Cross-browser responsiveness			
Phase 3: Stabilization & Testing	Sprint 9	Integrate all modules together	1/17/2026	1/30/2026	13
		Test full learner flow end-to-end			
		Prepare demo content for beta			
	Sprint 10	Unit testing & bug fixing	1/31/2026	2/13/2026	13
		Database optimization			
		Code cleanup & documentation			
	Sprint 11	Beta deployment (Vercel/Heroku)	2/14/2026	2/27/2026	13
		Simulated learner testing			
		Feedback collection & small adjustments			
	Sprint 12	Final polish of UI/UX	2/28/2026	3/12/2026	12
		Test edge cases (retry, rewards, payments)			
		Prep screenshots/video for report			
	Sprint 13	Load final micro-skill content	3/13/2026	3/26/2026	13
		Ensure all lessons, quizzes, and rewards work			
		Internal integration testing			
	Sprint 14	Performance tuning (loading speed, DB queries)	3/27/2026	4/9/2026	13
		Minor bug fixes			
		Demo rehearsal for internal defense			
	Sprint 15	Full dry-run of system	4/10/2026	4/23/2026	13
		Internal review with supervisor			
		Fix any critical issues			
	Sprint 16	Launch-ready system finalization	4/24/2026	4/30/2026	6
		Backup, deployment, final report prep			
		Internal defense demo			

FIG: PROJECT TIMELINE

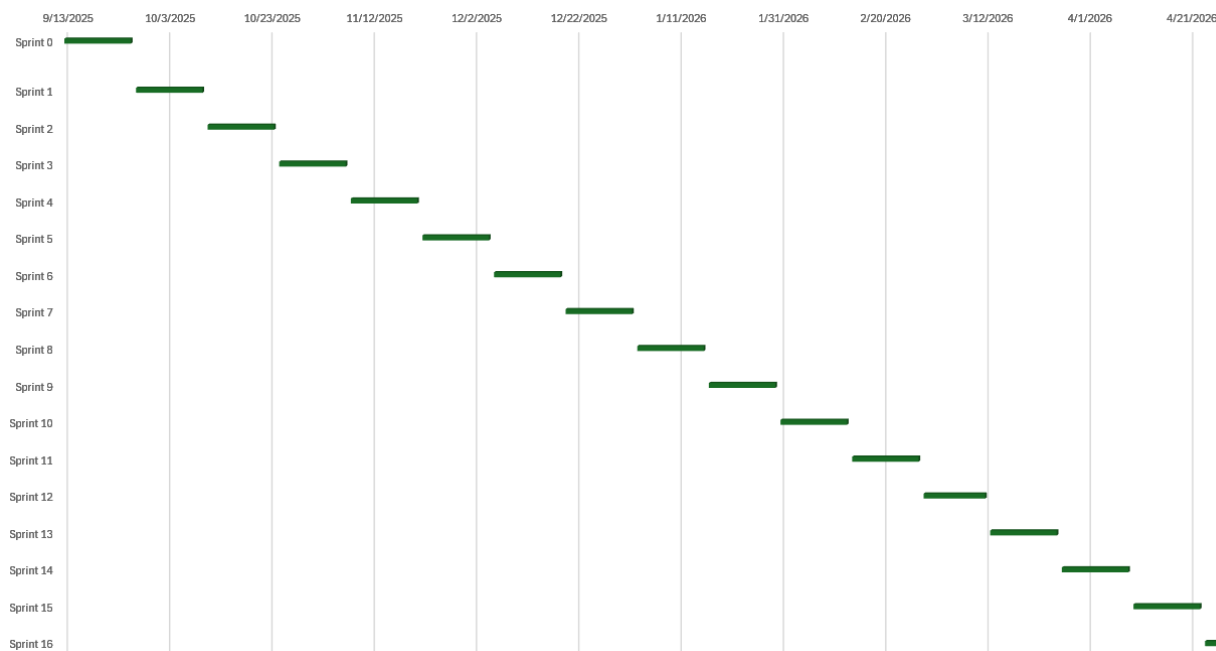


FIG: GANTT-CHART

Risk Management

Risk Id	Risk Description	Category	Likelihood	Impact	Priority	Mitigation Strategy	Contingency Plan
R01	Scope creep - adding too many features	Project Management	High	High	Critical	Define clear MVP scope, use Agile sprints to manage changes, strict feature prioritization	Cut non-essential features, extend timeline if approved by supervisor
R02	Timeline delays due to technical complexity	Technical	Medium	High	High	Build buffer time into sprints, prioritize core features first, weekly progress reviews	Reduce feature scope, focus on MVP, request deadline extension
R03	Backend API failures or bugs	Technical	Low	Critical	High	Unit testing with Jest, API testing with Postman, code reviews	Implement robust error handling, maintain backup database snapshots
R04	Database connection issues or data loss	Technical	Low	Critical	High	Use MongoDB Atlas for reliability, implement regular backups, test connection handling	Switch to backup database instance, restore from recent backup
R05	Authentication security vulnerabilities	Security	Medium	Critical	Critical	Use bcrypt for password hashing, implement JWT properly, security testing	Immediate patching, review auth flow, consult security resources
R06	Frontend-backend integration problems	Technical	Medium	Medium	Medium	Regular integration testing, use consistent API contracts, Postman collection for testing	Debug systematically, check CORS settings, verify endpoint connections
R07	Poor user interface/experience	Design	Medium	Medium	Medium	User feedback sessions, usability testing with 5-10 users, iterative design improvements	Redesign problem areas based on feedback, consult UI/UX resources
R08	Lack of engaging gamification elements	Design	Medium	High	High	Research best practices, test with users, iterate based on feedback	Simplify gamification, focus on core mechanics (points, badges)
R09	Payment gateway integration failures	Technical	Medium	High	High	Use Stripe test mode extensively, follow documentation carefully, handle edge cases	Use simpler payment stub, mark as future enhancement
R10	Insufficient content (micro-skill lessons)	Content	High	High	Critical	Write 2-3 lessons per week starting early, reuse lesson templates	Reduce initial lesson count to 10-15, add more post-launch
R11	Figma AI generates unusable code	Technical	Low	Medium	Low	Review and test all generated code, modify as needed, use as starting point only	Code components manually, use Figma only for design inspiration
R12	Server hosting/deployment issues	Infrastructure	Medium	Medium	Medium	Deploy early to staging environment, use reliable platforms (Vercel, Railway)	Switch to alternative hosting platform, use local demo for defense
R13	Browser compatibility issues	Technical	Low	Low	Low	Test on major browsers (Chrome, Firefox, Safari, Edge), use standard web technologies	Document supported browsers, provide fallback UI
R14	Literature review quality insufficient	Academic	Low	Medium	Medium	Use recent peer-reviewed sources (2020+), properly cite all sources, get supervisor feedback	Add more sources, expand analysis, revise based on feedback
R15	Supervisor unavailability or miscommunication	Project Management	Medium	Medium	Medium	Schedule meetings in advance, send progress updates via email, document decisions	Proceed with best judgment, document reasoning, catch up at next meeting
R16	Burnout or time management issues	Personal	Medium	High	High	Realistic sprint planning, take regular breaks, maintain work-life balance	Reduce daily work hours, extend sprint length, seek support
R17	COVID-19 or health issues causing delays	External	Low	High	Medium	Maintain good health practices, have work-from-home setup ready	Notify supervisor immediately, adjust timeline, work remotely
R18	Technology stack changes mid-project	Technical	Low	High	Medium	Commit to chosen stack early, only change if critical issue arises	Assess impact, minimize changes to affected modules only
R19	Data validation and input sanitization gaps	Security	Medium	Medium	Medium	Implement input validation on both frontend and backend, test with invalid data	Add validation middleware, sanitize all user inputs
R20	Mobile responsiveness issues	Design	Medium	Low	Low	Tailwind responsive classes, test on multiple screen sizes regularly	Focus on desktop first, mark mobile as future work

Risk Register (Narrative Format)

This section provides a summarized explanation of each risk recorded in the register. Each entry includes the risk ID, a brief description, its category, likelihood level, impact level, overall priority, and both mitigation and contingency strategies.

R01 – Scope creep

Too many features might be added beyond the MVP. It falls under project management with high likelihood and high impact, making it critical. Mitigation focuses on defining a clear MVP and using Agile to manage scope. If things go wrong, non-essential features will be cut.

R02 – Timeline delays

Complex technical tasks could push deadlines. Likelihood is medium but impact is high, so it's a high priority risk. The plan is to build buffer time and review weekly. If delayed, the scope will be reduced or a deadline extension will be requested.

R03 – Backend bugs / API failure

APIs might crash or return errors. It has medium likelihood and high impact. Testing with Jest/Postman and code reviews will prevent this. Backup database snapshots are the backup strategy.

R04 – Database connection failure / data loss

Low likelihood but **critical impact**, so still high priority. Using MongoDB Atlas and backups helps prevent it. If it occurs, the project switches to a backup database.

R05 – Authentication security issues

Security is critical here. Medium likelihood but critical impact = critical priority. Proper JWT + bcrypt + testing are used to mitigate. If spotted, immediate patching and auth review will be done.

R06 – Frontend/Backend integration issues

APIs may not connect properly. This has medium likelihood and medium impact. Integration testing and consistent API contracts are used. If issues happen, debugging and endpoint verification will solve it.

R07 – Poor UI/UX

Medium risk in both likelihood and impact. User testing and iterative design will help. If feedback is negative, UI sections will be redesigned.

R08 – Weak gamification elements

Users might not feel engaged. Medium likelihood but high impact, so it's high priority. Researching best practices and testing with users helps. If required, features will be simplified (basic points/badges only).

R09 – Payment gateway failure

Payment might fail or return errors. Medium likelihood and high impact → high priority. Stripe test mode and documentation help avoid this. If it fails, a simpler payment stub will replace it.

R010 – Insufficient content (lessons)

A high risk: high likelihood + high impact → critical. The plan is to write lessons weekly using templates. If delays occur, only 10–15 core lessons will be launched first.

R011 – Bad Figma AI code generation

Figma AI may output useless code. Low likelihood and medium impact → low priority. Code must be reviewed and used only as a starting point. If the AI fails, components will be made manually.

R012 – Server deployment issues

Hosting might fail during deployment. Medium likelihood + medium impact = medium priority. Deploying early on staging prevents surprises. If that fails, a local demo will be used.

R013 – Browser compatibility issues

Some browsers may not render correctly. Likelihood and impact are both low → low priority. Testing on major browsers is planned. If unsupported, fallback UI will be documented.

R014 – Poor literature review quality

Research may not meet academic standard. Low likelihood but medium impact. Using recent sources and supervisor feedback helps. If quality is low, more sources will be added and analysis expanded.

R015 – Supervisor unavailable

Risk of delayed communication. Likelihood and impact are both medium. Meetings will be scheduled early and updates emailed. If unavailable, you'll proceed using best judgment and record decisions.

R016 – Burnout / time management issues

A real threat with medium likelihood but high impact, making it high priority. Realistic sprint planning + rest is mitigation. If burnout happens, work hours will be reduced.

R017 – Health/COVID delays

Low likelihood but high impact. Maintaining health and having work-from-home setup are preventive. If it happens, remote progress continues and supervisor is notified immediately.

R018 – Tech stack changes mid-project

Switching technologies could blow up the timeline. Likelihood low, impact high → medium priority. Stack will be finalized early. If forced to change, only affected modules get modified.

R019 – Weak validation / sanitization

Risk of insecure user input. Medium likelihood and medium impact. Validation on both backend and frontend is planned. If missed, validation middleware will be added later.

R020 – Mobile responsiveness issues

App may look bad on smaller screens. Medium likelihood but *low impact* → low priority. Tailwind responsive classes will be used. Worst case: desktop-first delivery, mobile marked as future update.

Limitations

While this project aims to provide an engaging and modular platform for technical micro-skills, several limitations must be acknowledged:

1. **Scale of Evaluation:** The prototype will undergo only informal usability testing with a small group of participants. As a result, quantitative data on engagement, retention, or learning effectiveness will not be collected at this stage.
2. **Limited Personalization:** Advanced AI-driven adaptivity and personalized learning paths are out of scope, which may limit effectiveness for learners with highly diverse skill levels or learning styles.
3. **Platform Reach:** The current implementation is web-based only. Mobile app support and offline accessibility are not included in this iteration.
4. **Content Scope:** The platform focuses on a select set of micro-skills (SQL basics, Excel formulas, networking fundamentals, cybersecurity awareness). Expansion to other technical areas will require future work.
5. **Resource Constraints:** Time, software licensing, and availability of assets may constrain the depth and complexity of gamification features or interactive animations.

Despite these limitations, the prototype provides a proof-of-concept demonstrating the potential of gamified microlearning, laying the foundation for further enhancements and broader studies. These limitations guide the scope of the MVP and ensure realistic delivery within project constraints

Conclusion & Future Scope

Conclusion:

This project successfully designed and developed a web-based gamified learning platform for technical micro-skills, implementing a modular structure for lessons, interactive feedback, and a rewards system. The platform allows learners to progress through micro-skill modules, track their achievements, and engage with content in a gamified manner, fulfilling the objectives outlined in the project proposal. Core functionalities, including user authentication, lesson modules, difficulty selection, progress tracking, and basic payment flow for advanced levels, have been implemented and tested. The system meets the minimum requirements for FYP submission.

Future Scope:

While the platform meets the minimum requirements for a launch-ready MVP, several enhancements can be considered for future development:

1. **Mobile Application Development:** Extend the platform to iOS and Android devices to increase accessibility.
2. **Adaptive Learning & AI Integration:** Implement AI-driven content recommendations and adaptive difficulty adjustment for personalized learning experiences.
3. **Enhanced Analytics:** Provide in-depth dashboards for administrators to analyze learner performance and lesson effectiveness.
4. **Advanced Gamification:** Introduce leaderboards, social features, and collaborative learning challenges to boost engagement.
5. **Expanded Micro-skill Catalog:** Continuously add new technical micro-skills, covering a broader range of domains to cater to diverse learner needs.

These future enhancements aim to improve engagement, scalability, and personalization, ensuring that the platform can evolve into a more comprehensive learning ecosystem beyond the current MVP.

Research Articles & Studies

1. Caponetto, I., Earp, J. and Ott, M., 2014. Gamification and education: A systematic review of literature. *Journal of e-Learning and Knowledge Society*, 10(3), pp.1–14.
2. Hamari, J., Koivisto, J. and Sarsa, H., 2014. Does gamification work? A literature review of empirical studies on gamification. *Proceedings of the 47th Hawaii International Conference on System Sciences*, pp.3025–3034.
3. Wang, A.I., Zhu, M. and Sun, Z., 2018. Gamified mobile learning: Effects on student engagement and performance. *Computers & Education*, 125, pp.75–88.
4. Huang, W., Martinez-Maldonado, R. and Anderson, E., 2023. Personalized gamification in online higher education: Impact on engagement and persistence. *Journal of Educational Technology & Society*, 26(2), pp.34–49.
5. Salas-Morera, L., Redel-Macías, M. and Cubero, J., 2024. Gamification for technical micro-skills training: Engagement and retention outcomes. *International Journal of Educational Technology*, 21(1), pp.12–29.
6. Sailer, M. and Homner, L., 2020. The gamification of learning: A meta-analysis. *Educational Research Review*, 30, pp.1–15.
7. Landers, R.N. and Armstrong, M.B., 2017. Enhancing instructional outcomes with gamification: A systematic review. *Journal of Management Education*, 41(6), pp.709–739.
8. Alhammad, M. and Moreno, R., 2018. Adaptive learning in gamified systems: Current challenges. *Computers & Education*, 126, pp.123–136.

9. Kumar, S., Lee, J. and Alvarez, D., 2024. Designing gamified microlearning for digital-native learners. *Learning Technologies Journal*, 35(1), pp.45–67.
10. Caballé, S. and Clarisó, R., 2014. Balancing scalability and interactivity in online learning platforms. *International Journal of Web-Based Learning*, 9(2), pp.55–70.
11. Alhammadi, R., Wallace, T. and Moreno, S., 2023. AI-powered adaptivity in gamified learning systems. *Journal of Educational Computing Research*, 61(5), pp.1223–1245.

Tools, Libraries, & Platforms

Category	Tools / Platforms	Links (if applicable)
Frontend	React.js, Framer Motion, Lottie	https://reactjs.org / https://www.framer.com/motion / https://airbnb.io/lottie
Backend	Node.js, Express.js	https://nodejs.org / https://expressjs.com
Database	MongoDB	https://www.mongodb.com
Game Engine	Phaser.js, Unity WebGL	https://phaser.io / https://unity.com/webgl
Design & Assets	Figma, Illustrator, Blender, OpenGameArt.org, Freesound.org	https://www.figma.com / https://www.adobe.com/products/illustrator.html / https://www.blender.org / https://opengameart.org / https://freesound.org
Animation	Lottie, Framer Motion	https://airbnb.io/lottie / https://www.framer.com/motion
UML Diagrams	Figma Workspace	https://www.figma.com/design/YV9V3l6Euxtl46BIneuWMG/FYP-diagrams?node-id=0-1&p=f&t=6e7LNydF0RDdzYNP-0